

Priyanka Chatterjee

Python + AWS Backend Engineer · REST APIs & Distributed Systems · 9,300 ★ OSS

Contributor · 3.5+ Years

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SUMMARY

Python + AWS Engineer with 3.5+ years of experience and expertise in **Python** backend development, **AWS cloud-native** architecture (Lambda, EC2, S3), RESTful API design, and **distributed systems**. Led an **AWS** SDK migration across 180+ call sites and 11+ **AWS** services at Turbot HQ, keeping production systems supported past **AWS** SDK v2 end-of-life while cutting policy-code duplication ~60%.

TECHNICAL SKILLS

Languages: Python, Go, Java, TypeScript, JavaScript (ES6+), SQL, Bash

Backend & APIs: REST, API Security, gRPC, OAuth 2.0 (Auth0), JWT, Microservices, Node.js, Express.js, Gin

AI Engineering (Python): LLM Application Development, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Pydantic AI, OpenAI Agents SDK

Cloud & Serverless (AWS-focused): AWS (Lambda, EC2, RDS, S3, SecurityHub, GuardDuty, ControlTower, Shield, X-Ray, Comprehend, Pinpoint, RAM, EFS), Azure, GCP

Distributed Systems & Messaging: Distributed Systems, Event-Driven Architecture, Apache Kafka (consumer groups, DLQs), AWS SNS/SQS, AWS EventBridge

Databases & SQL: PostgreSQL, MySQL, DuckDB, MongoDB, Redis; jsonb, CTEs, query tuning, prepared statements

DevOps & Testing: Docker, GitHub Actions, Git, Unit Testing, Integration Testing, Code Review

WORK EXPERIENCE

Turbot HQ, Inc. Remote

August 2022 – April 2026

Software Engineer I

- Built the Steampipe Model Context Protocol (MCP) server end to end in **Python**, giving LLM agents standardized, real-time access to enterprise cloud and **security** data for retrieval and analysis.
- Built a RAG-based internal Pydantic AI agent in **Python** to resolve cross-team queries, ingesting internal documentation into S3 vector storage; deployed it as a **serverless AWS** Lambda function and integrated it with the Slack Bot API for instant developer access.
- Designed and developed an internal **REST** API microservice in Go using the Gin framework to power the backend of Turbot Guardrails, a **cloud-native** SaaS platform; secured endpoints with OAuth 2.0 authentication via Auth0, containerized with Docker, deployed to **AWS** EC2, and served as a gRPC client for several internal microservices.
- Built a Kafka-based, event-driven distributed pipeline where consumers asynchronously processed policy-enforcement events and fired a webhook once a policy was successfully enforced; added a dead-letter queue (DLQ) to capture recurring failures for debugging.
- Led the **AWS** SDK for JavaScript v2 to v3 migration across 180+ call sites and 11+ **AWS** services (SecurityHub, CloudWatch, ControlTower, Shield, Comprehend, Pinpoint, RAM, X-Ray, EC2, EFS) in production Lambdas; reduced policy-code duplication ~60% via templating, keeping the platform supported past **AWS** SDK v2 EOL.
- Optimized complex PostgreSQL queries powering customer compliance audits — rewriting **AWS** IAM policy-evaluation queries with jsonb array elements and multi-stage CTEs — improving query performance 40% and eliminating a class of false-positive findings.

PROJECTS

General-Purpose AI Chatbot / Agent (Model Context Protocol) (Jun 2025 – Aug 2025)

Agentic chatbot/agent framework built in **Python** with Pydantic AI featuring configurable external MCP servers via a persistent MongoDB-backed registry, JSON config loader with strict input validation, per-server failure isolation, native tool calling, mid-conversation multi-model switching (OpenAI, Gemini, Anthropic, Mistral), conversation-history persistence, sync + streaming output, and Logfire observability.

CIS Benchmark → Compliance Dashboard Generator

Python CLI that automates compliance-dashboard authoring for a multi-cloud governance product. Ingests CIS benchmark XLSX workbooks (**AWS**, Azure) and emits ready-to-deploy HCL dashboard config plus markdown docs. OOP resource model, custom HCL pretty-printer, openpyxl + pandas, distributed as a pip-installable package.

EDUCATION

University of Engineering & Management (UEM) Kolkata, WB, IN

August 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 9.28

Relevant Coursework: Object-Oriented Programming, DBMS, Data Structures and Algorithms, Shell Scripting in Linux, Low-Level Design, High-Level Design